

Unit V: Fourier Transforms

Class 2:

- Find the Fourier transform of the slit function $f(t)$ defined as

$$f(t) = \begin{cases} \frac{1}{\varepsilon} & |t| \leq \varepsilon \\ 0 & |t| > \varepsilon \end{cases}$$

$$\text{Ans: } \frac{2}{iu\varepsilon} \sinh \omega\varepsilon, \frac{2}{i}$$

Determine the limit of this transformation as $\varepsilon \rightarrow 0$

- Find the Fourier transform of $e^{-|t|}$ and hence evaluate $\int_0^{\infty} \frac{\sin t}{1+t^2} dt$

$$\text{Ans: } \frac{2i\omega}{\omega^2 + 1}, \frac{-\pi}{e}$$
